

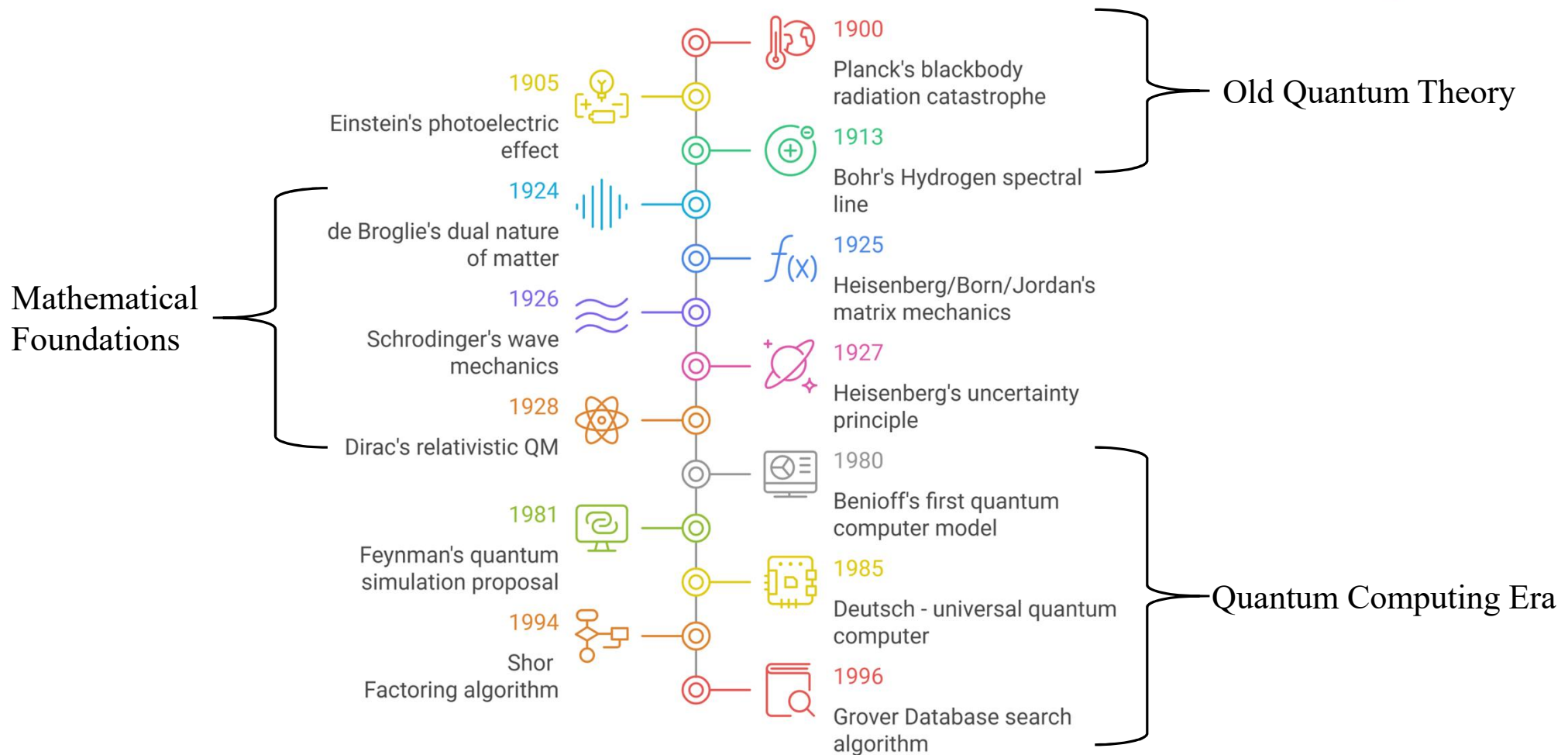
Quantum Computing From Qubits to Algorithms: An Exciting Learning and Career

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Outline

- Historical Perspectives
- Bringing Quantum Mechanical Features to Classical Computer Science
- Quantum Computing and its Building Blocks
- Visualizing and Simulating Quantum Circuits
- Quantum Algorithms
- Career Prospects

Milestones in Quantum Mechanics/Quantum Computing

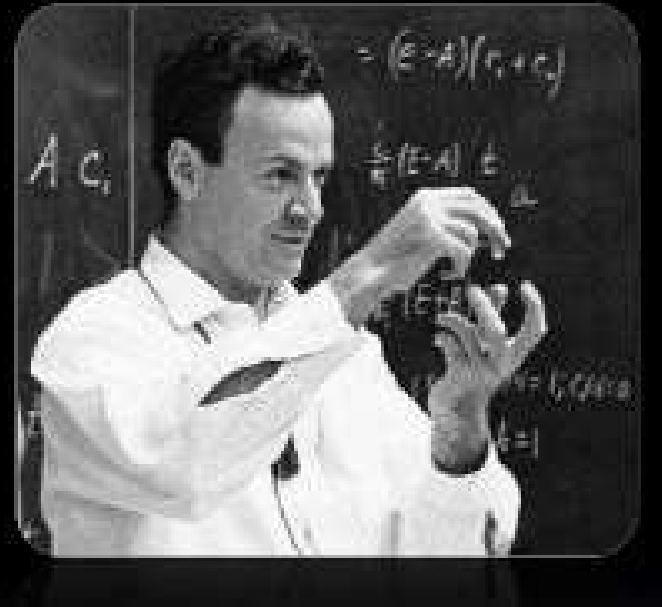


What is Quantum Mechanics?

General View (often misunderstood)	Corrected View
Quantum Mechanics is branch of Physics	It is not a branch, IT IS Physics!
It applies only at microscopic level	It applies at all scales even at astronomical scales
It gives probabilistic answers, answers involve uncertainties	Randomness and uncertainties are inherent in nature itself, QM models them
It is weird and not well understood	It is greatly understood. Mathematical part is well-defined; physical interpretations are different
It is only a matter of Physics and Chemistry	QM provides answers to all including working of ICs to space shuttles to crop growth and medicines ... everything!
It is of no use in Computer Science and IT/AI-DS	Future of Computer Science is Quantum Computing
So, what is Quantum Computing then?	

Feynman's Proposal of a Quantum Computer

1981 - Richard Feynman determines that it is impossible to efficiently simulate an evolution of a quantum system on a classical computer.



Build a machine which implements quantum mechanical effects and uses quantum mechanically prepared system (e.g. electron spin states)

What is a Quantum Computer?

Quantum computer is a machine that works on the principles of quantum mechanics and executes programs in the same manner as microscopic objects behave

It is based upon using Quantum Mechanical properties like

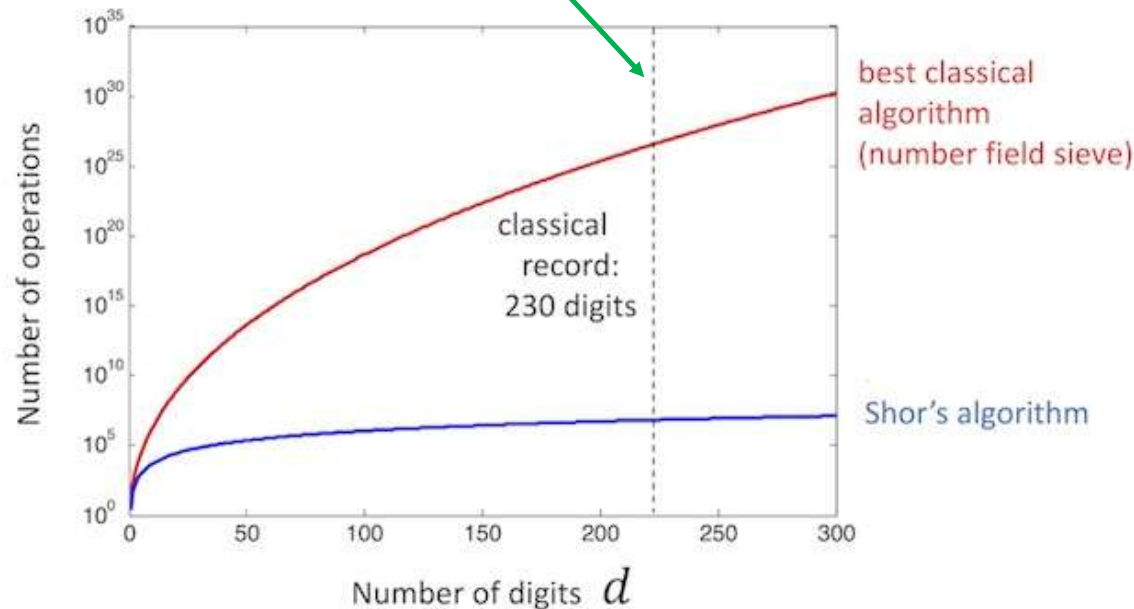
- Superposition/interference
- Entanglement/Bell States

For

- Improving computational powers manifold
- Reducing time and space complexity from exponential to polynomial rates

The “Quantum Supremacy”

≈ 75 CPU years (75 CPUs running continuously for 1 year)

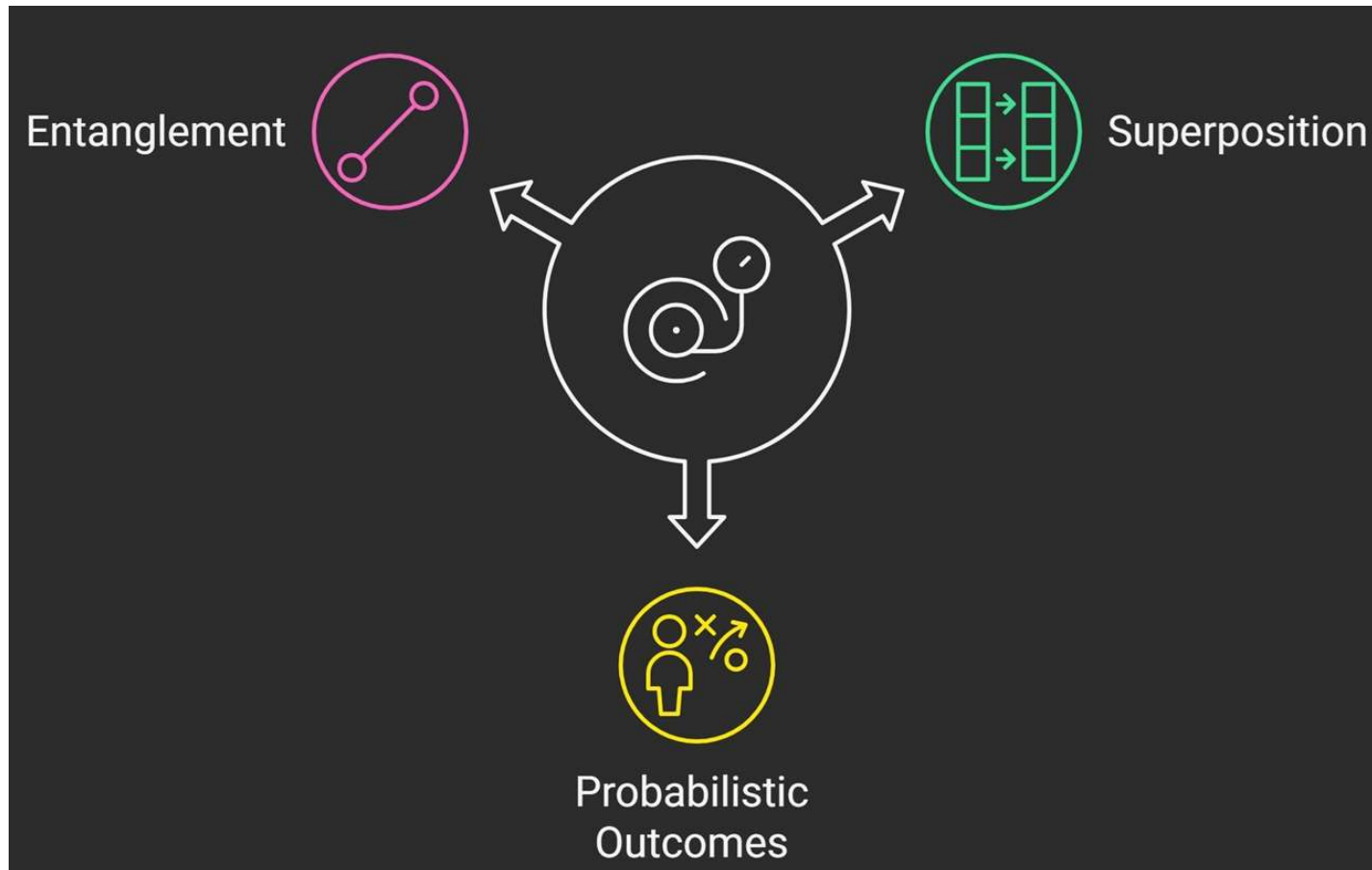


- 2001 IBM demonstrated Shor's algorithm by factorising number 15 using 7 qubits in a NMR system
- In 2012, factorized 21
- 2019, tried for 35

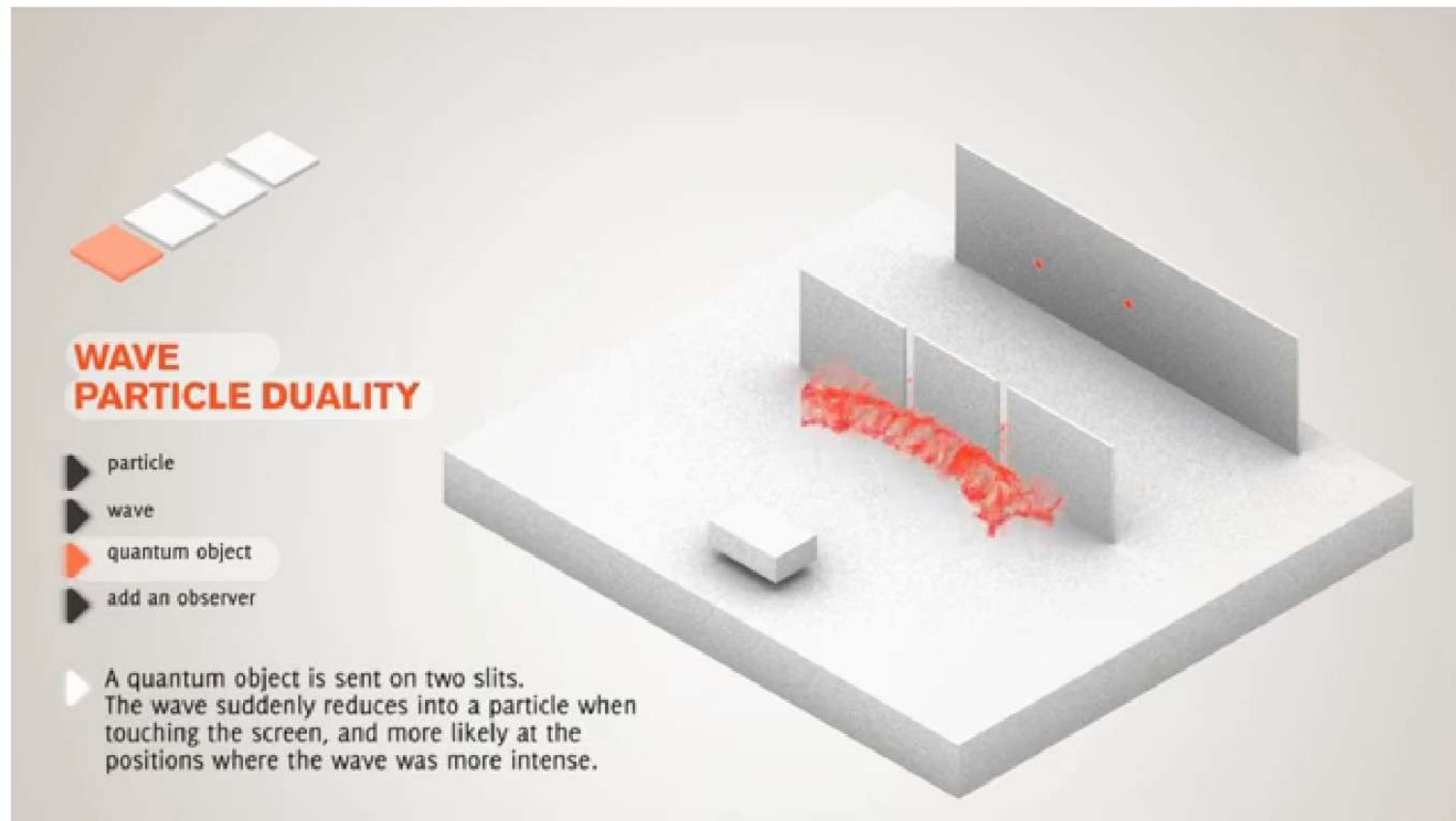
RSA230 (Rivest-Shamir-Adleman)

Graph credit: IBM Quantum Computing

Key Properties for attaining “Quantum Supremacy”

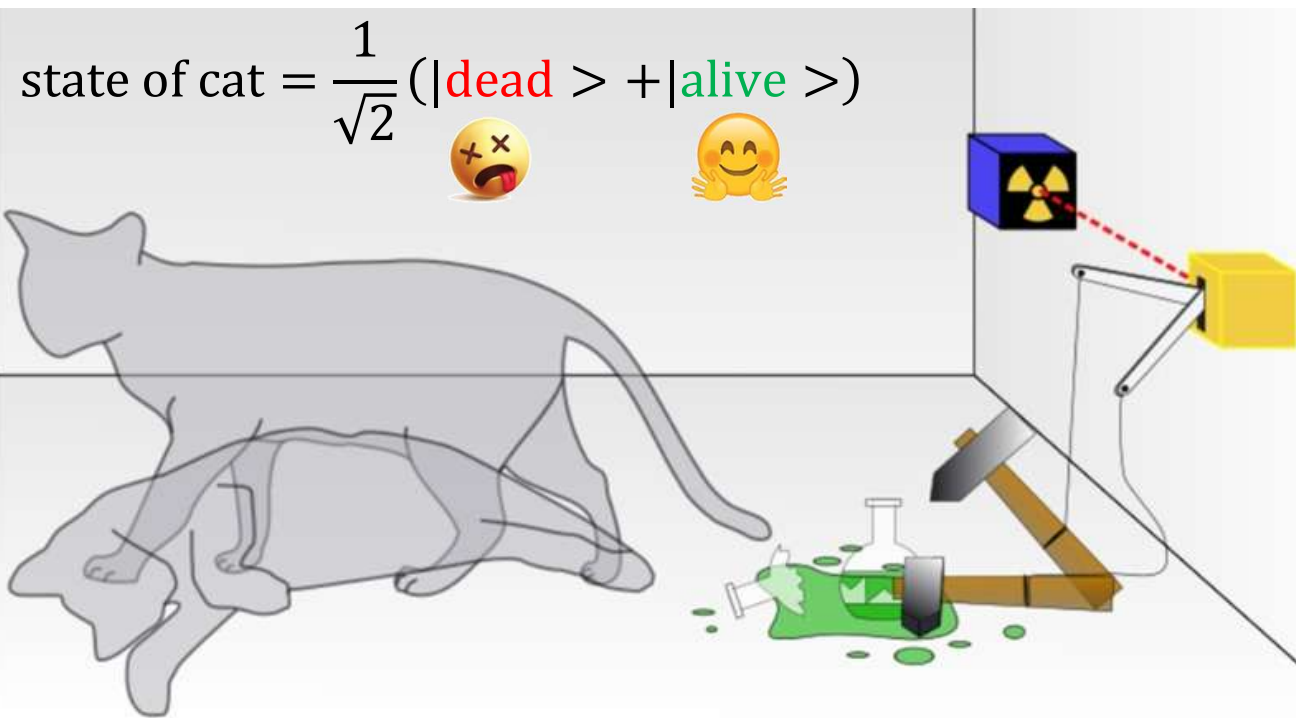


Quantum Effects – 1. Superposition/Wave-Particle Duality



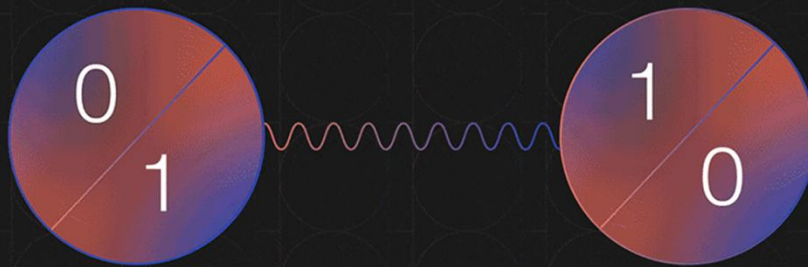
Quantum Effects – 2. Probabilistic Outcomes

Probabilistic result: $P = \int_a^b |\psi|^2 dx$; $\int_{-\infty}^{+\infty} |\psi|^2 dx = 1$



Quantum Effects – 3. Entanglement

Entanglement



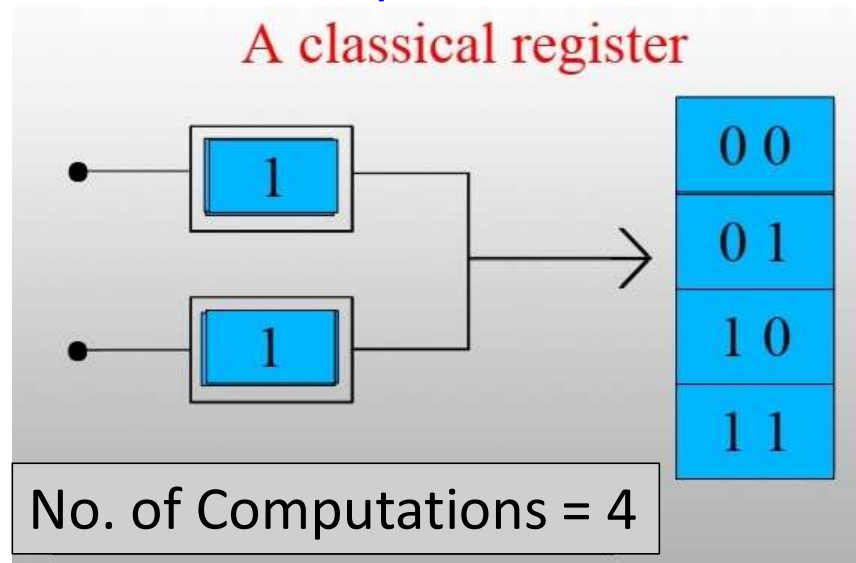
Object 1	Object 2	Entangled pair
0	0	00
0	1	01
1	0	10
1	1	11

$$|\varphi^{\pm}\rangle = \frac{1}{\sqrt{2}} (|00\rangle \pm |11\rangle)$$

Reason for The “Quantum Supremacy”

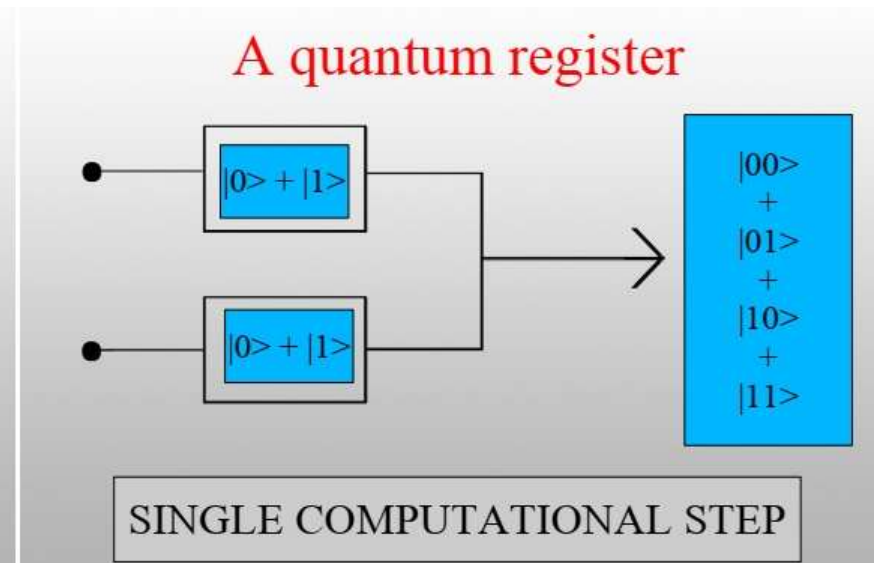
A classical computer

- Each register has unique input
- Executes one operation at a time



A quantum computer

- Each register has both inputs
- Executes all operations in one go

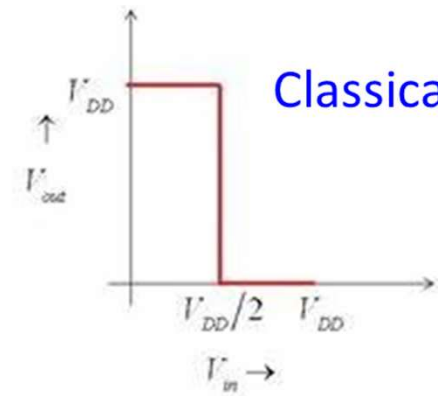
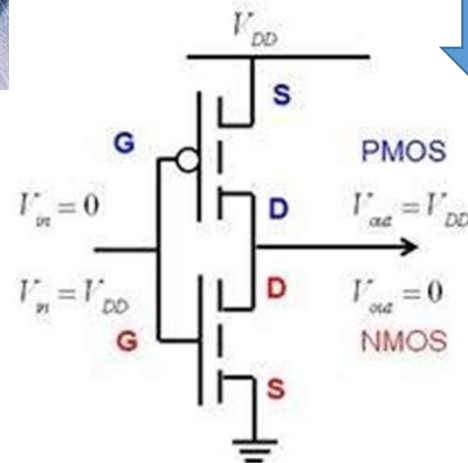
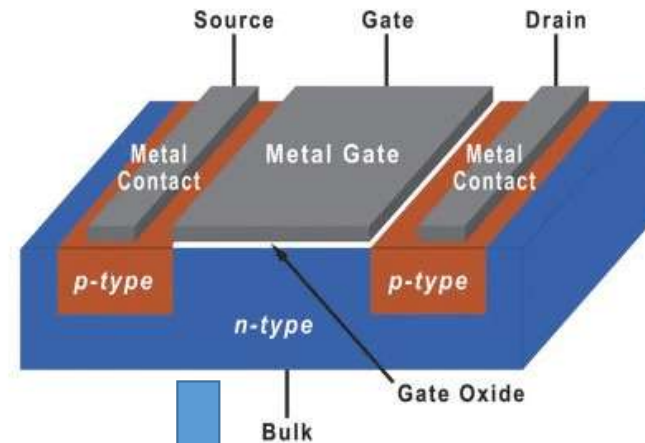
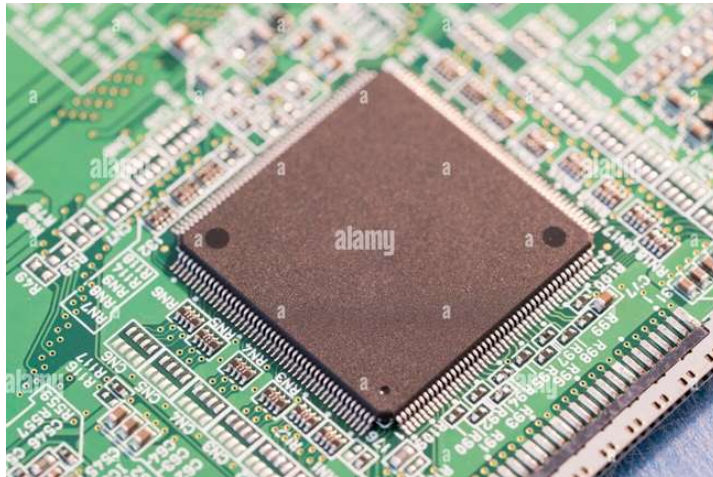


Classical v/s. Quantum Computers

Classical Computer	Quantum Computer
Uses semiconductor-based CMOS logic gates	Uses atomic, electronic, nuclear or photonic properties
High/Low voltage determines logic 1/0	Logic 1/0 represented by spin, electron state, light polarization etc.
Bit can be in state 1 or 0 at a given time	Qubits remain in superposed state before executing a command
Machine operates bit by bit	Machine operates on superposed qubits*

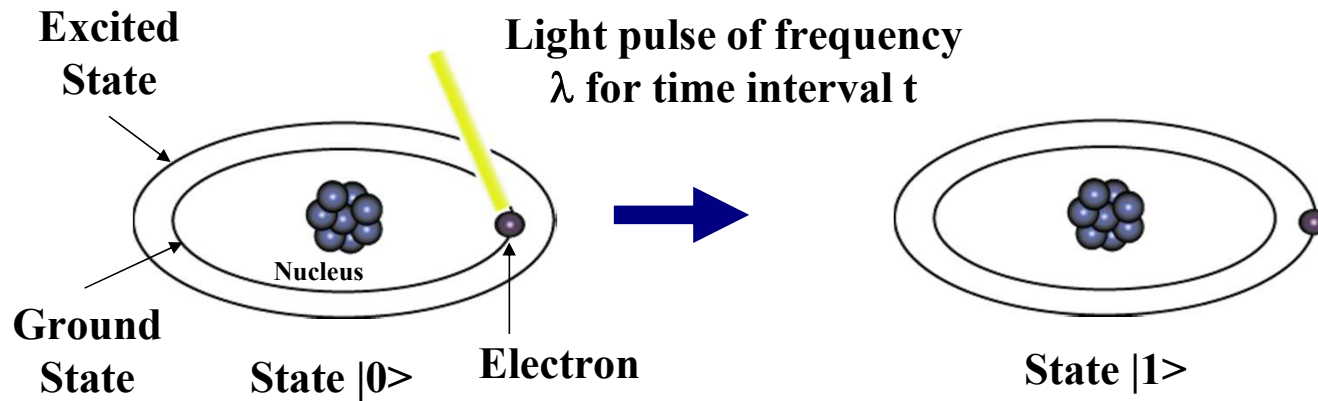
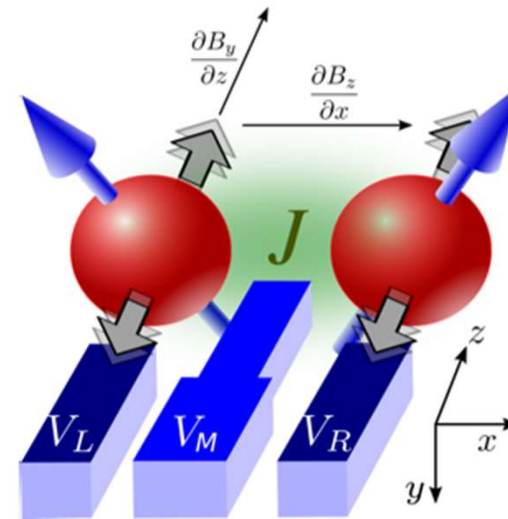
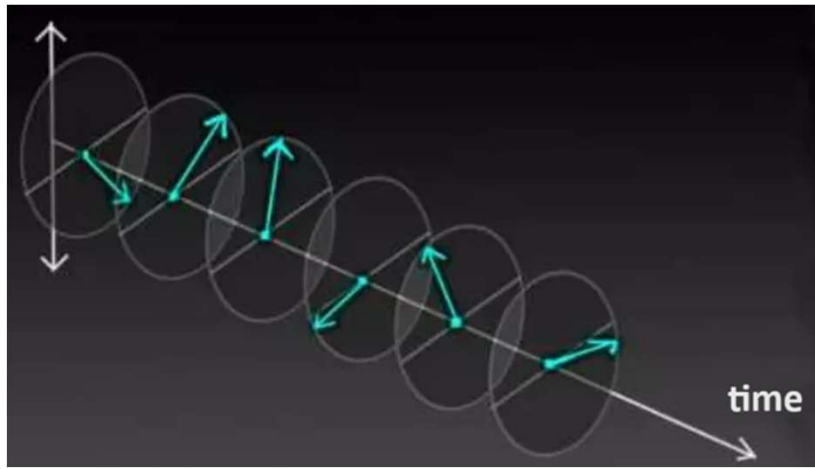
* This concept is different than parallel computing or super computing

Realization of Bits



Classical bits

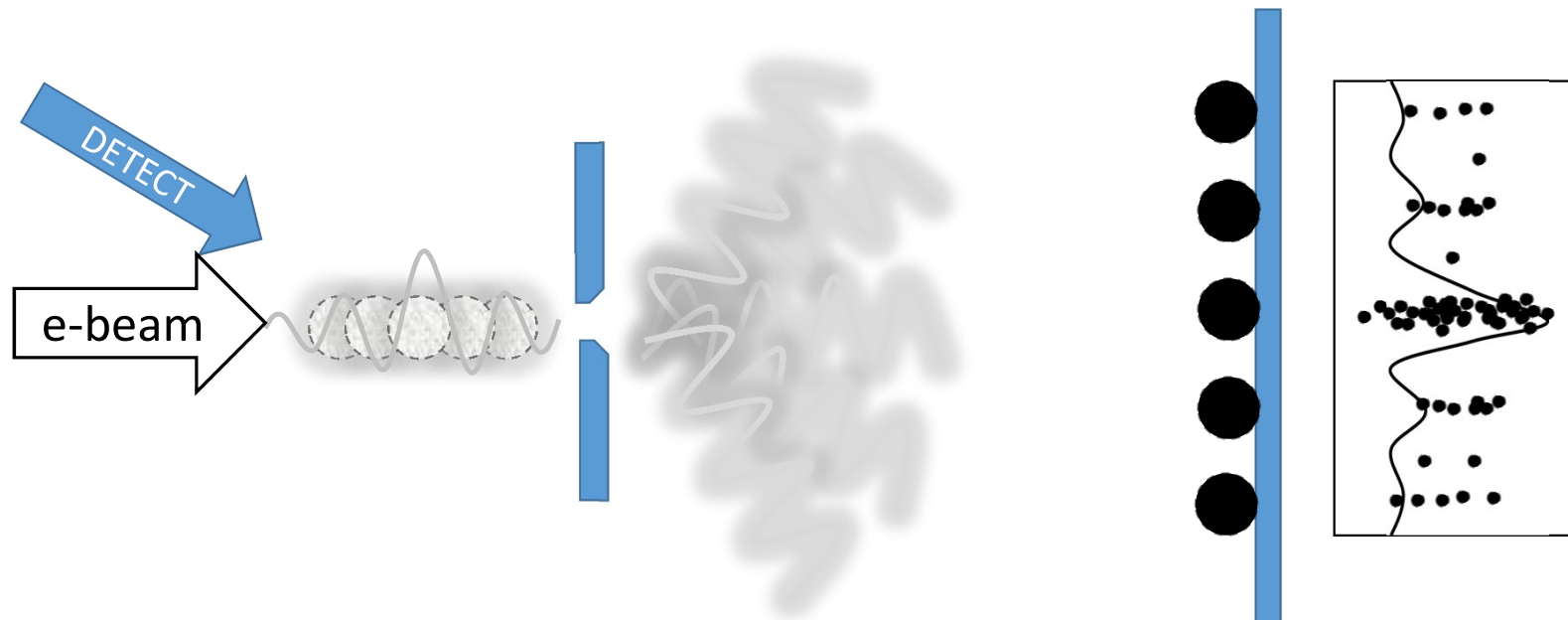
Realization of Qubits



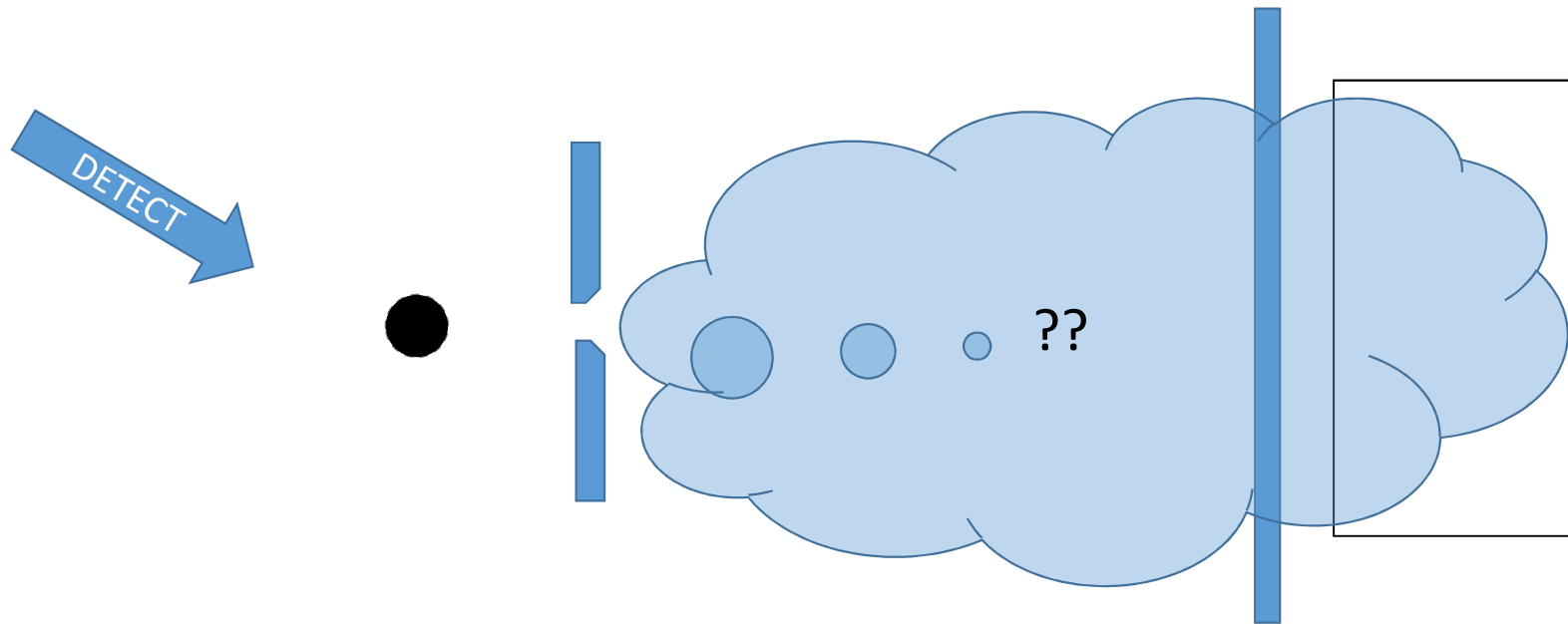
Hardware Options for Quantum Computers

Physics Concept	Qubits Realized using	Who is doing?
Ion trap	atomic energy levels	Honeywell, Samsung
SQUIDs	magnetisation	D-wave Technologies
NMR	nuclear spin	IBM
Quantum dots	Trapped electron energy states	INTEL, Diraq, ARQUE
Superconductors	Correlated electrons (Cooper pairs)	Google, IBM BARC

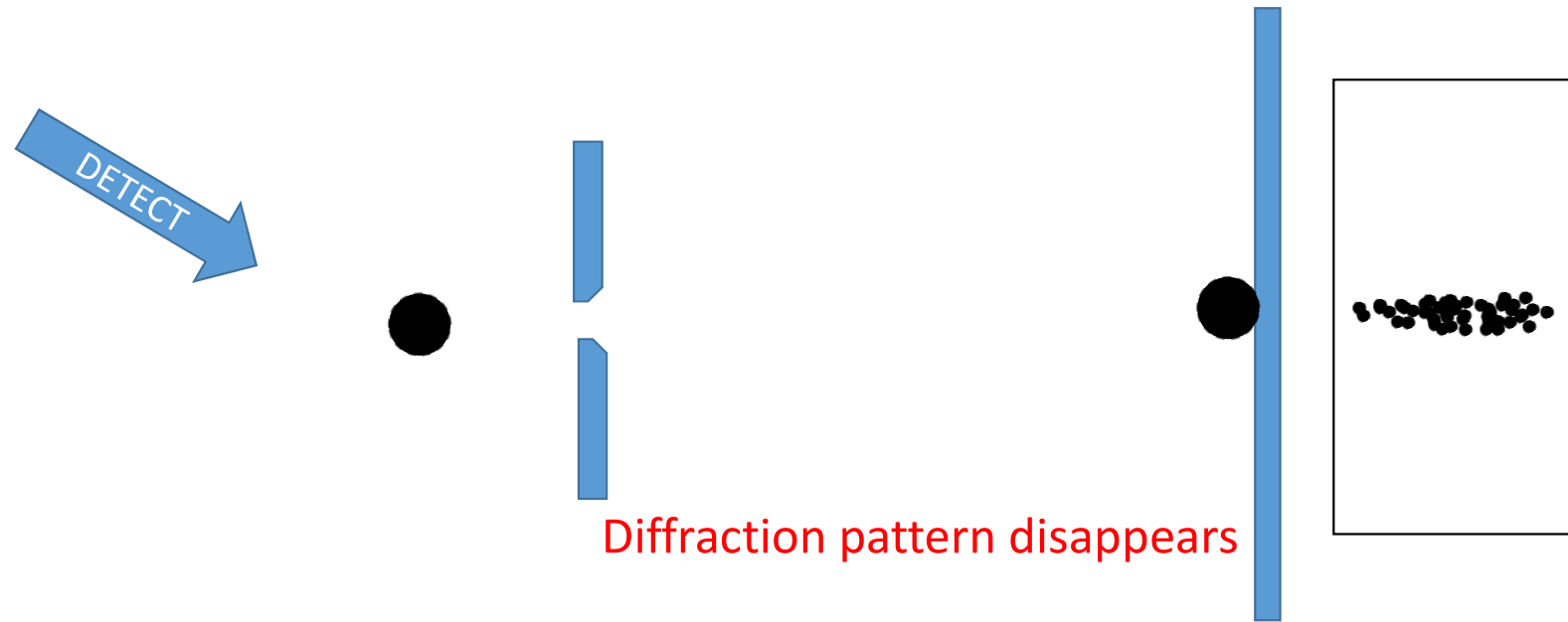
The Measurement Issue: Wave Function Collapse



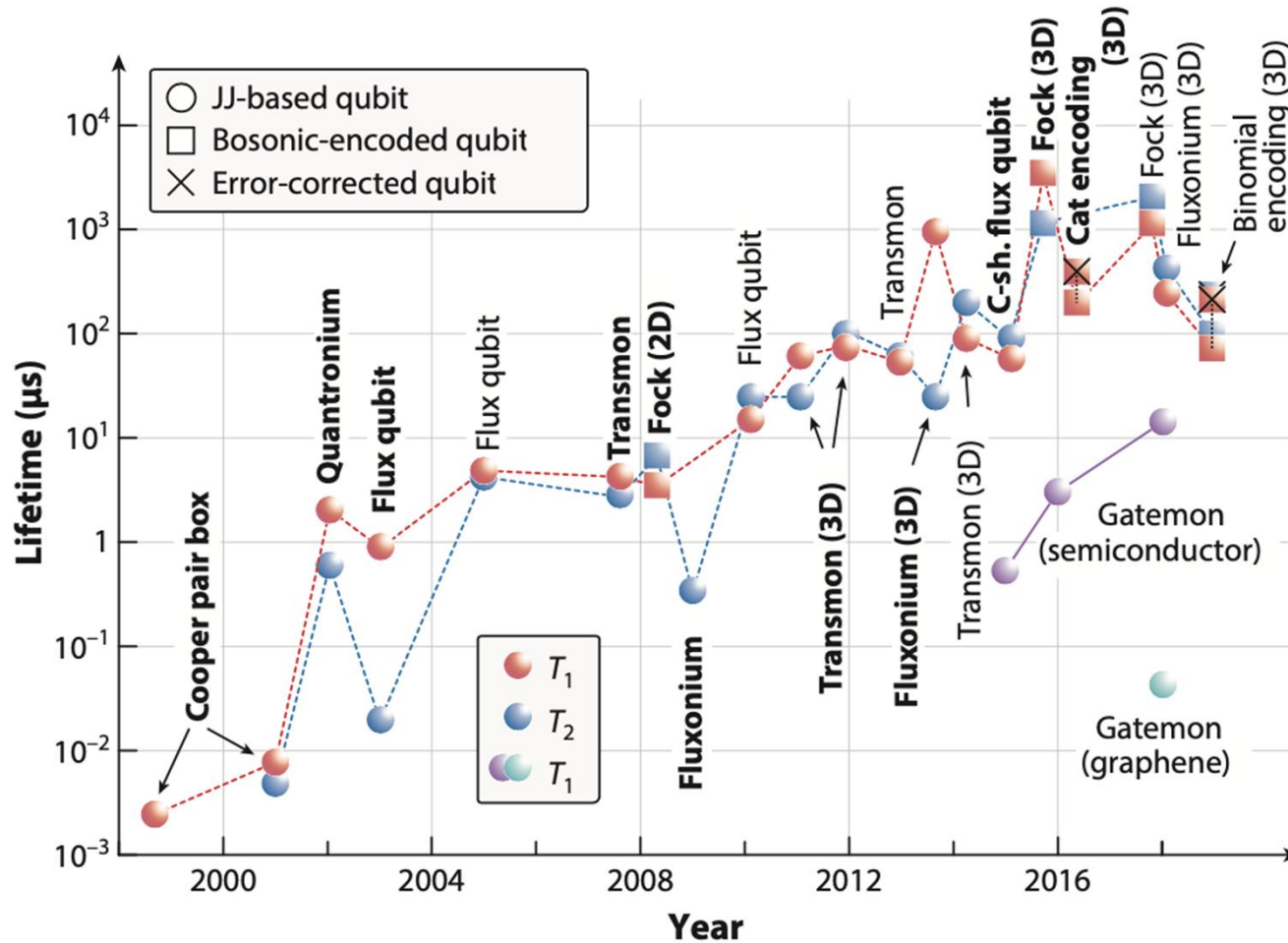
The Measurement Issue: Wave Function Collapse



The Measurement Issue: Wave Function Collapse



Holding the Superposed Qubit



Qubit Notations, Quantum Logic Gates, Quantum Circuits

Bits v/s Qubits

Bits

0 1
↑ ↑
States

$$b = \{0, 1\}$$

State is
Deterministic

Qubits

$|0\rangle$ $|1\rangle$
↑ ↑
States

(Unit Vectors)

$$b = \{0, 1, q\}$$

Complex Numbers
 $\alpha|0\rangle + \beta|1\rangle$
↑
Superposition

(Linear Combination)

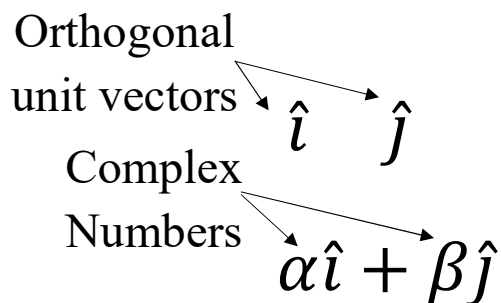
$$|q\rangle = \alpha|0\rangle + \beta|1\rangle$$

State is $|0\rangle: |\alpha|^2$
Probabilistic $|1\rangle: |\beta|^2$

Qubit Notations

Dirac Notations (Bra-Ket)

Vectors



ket $|0\rangle$ $|1\rangle$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

bra $\langle 0|$ $\langle 1|$

conjugate transpose of a ket

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \rightarrow \begin{bmatrix} \alpha^* & \beta^* \end{bmatrix}$$

ket

bra

(column vector)

(row vector)

Matrices (Linear Algebra)

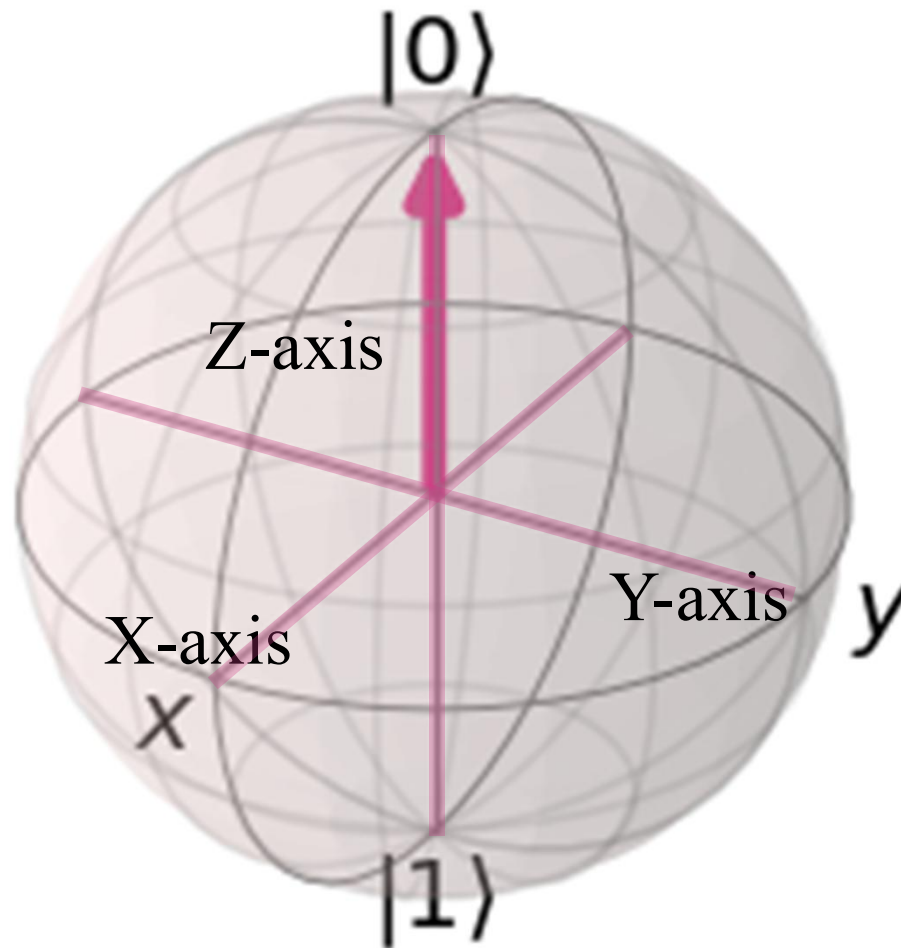
$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ Element in a complex
Hilbert (vector) space

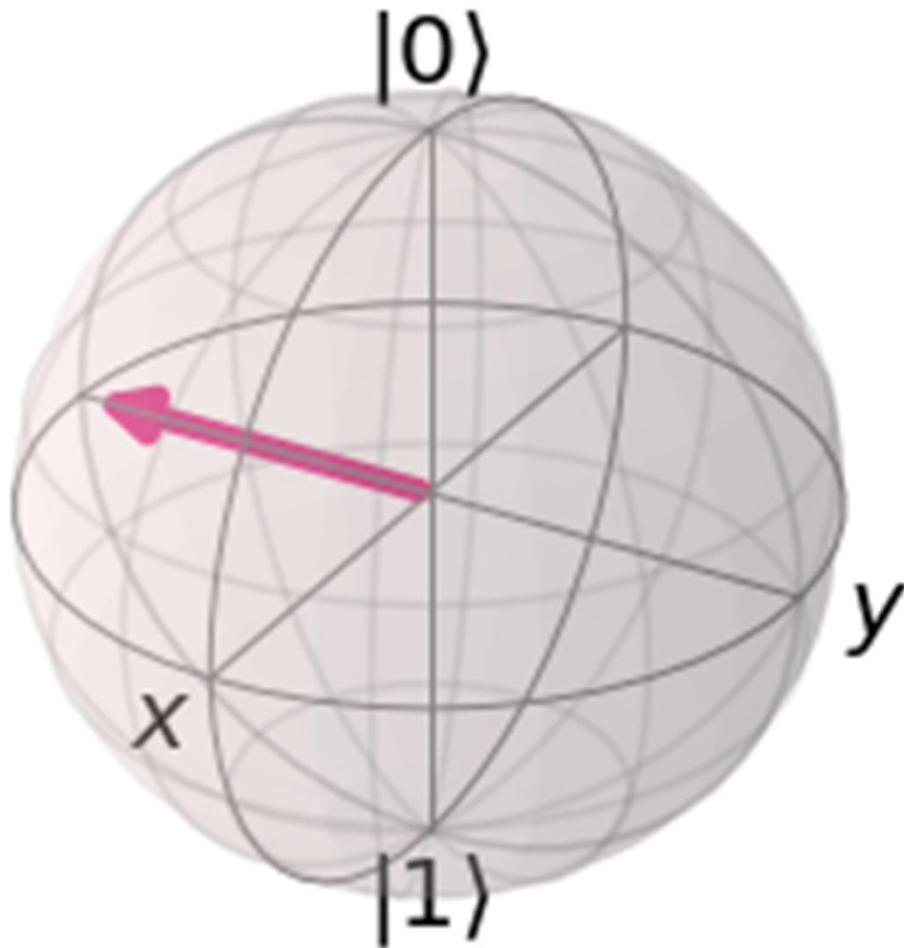
Linear combination of
orthonormal basis states

Visual Representation of Qubit States

Bloch Sphere



Bloch Sphere



$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

$$|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) = \begin{bmatrix} 1/\sqrt{2} \\ i/\sqrt{2} \end{bmatrix}$$

$$|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle) = \begin{bmatrix} 1/\sqrt{2} \\ -i/\sqrt{2} \end{bmatrix}$$

Quantum Circuits

Sequence of building blocks (gates)

Gates carry out elementary computations

Circuits carry out complex computations

q 

$$H\sigma_X = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$H\sigma_X|0\rangle = |- \rangle$$

$$H\sigma_X|1\rangle = | + \rangle$$

q 

$$SH = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$SH|0\rangle = | + i \rangle$$

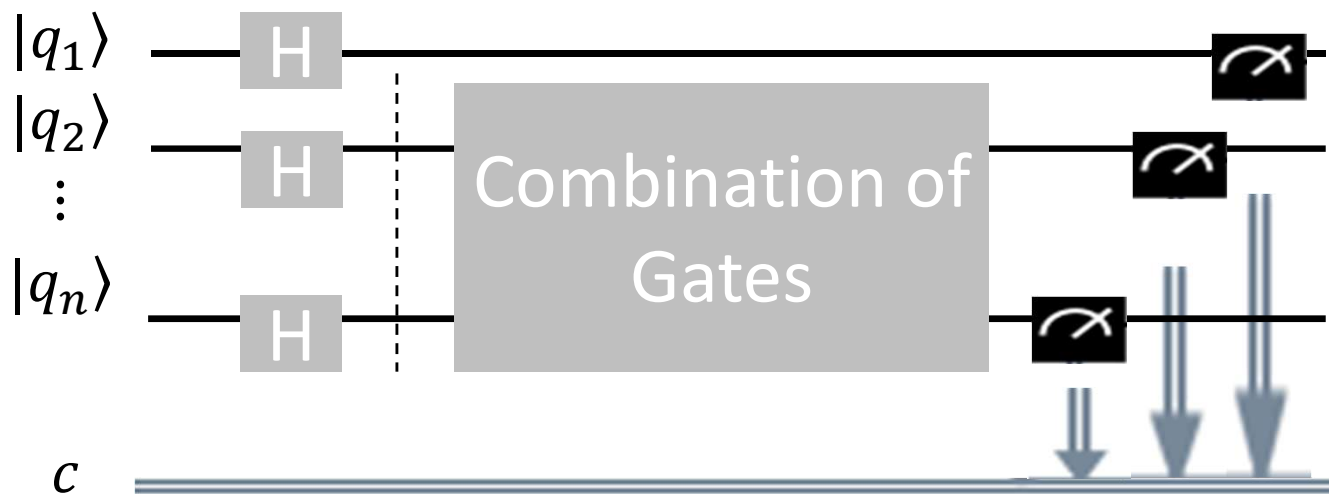
$$SH|1\rangle = | - i \rangle$$

Flow of a Quantum Circuits

Determine state space and inputs

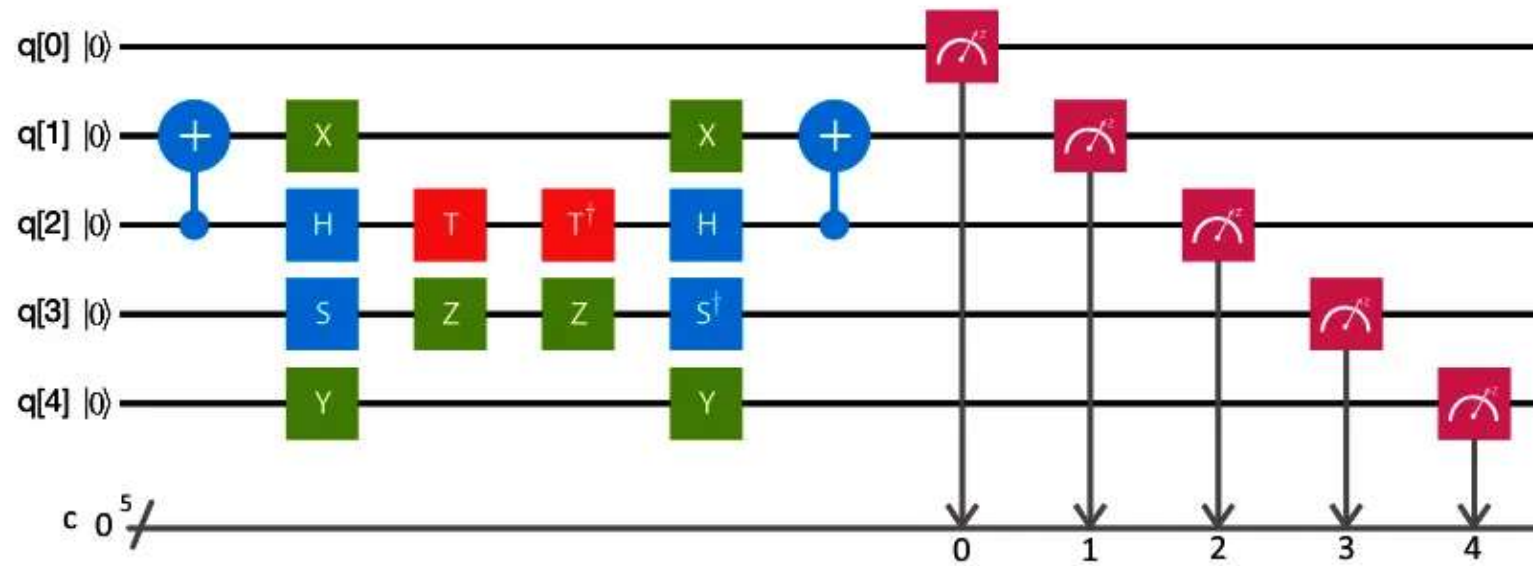
Obtain equal superposition of 2^n states

Perform transformations of superposition states, and use entanglement and interference effects to bias correct answers (states) and diminish incorrect ones



Measure outputs using a number of measurements for high accuracy

A sample Quantum Circuit



Simulating Quantum Circuits (IBM Qiskit)

quantum.cloud.ibm.com/composer?initial=N4lgjghgzgtiBcIDyAFAogOQloEEDKAsgAQBMAAdAAwDcAOgHYCWdAxdYCuAJgKZE3jdW...

scholar WebofScience Youtube LMS gmail Sign in | IBM Quant... Virtual Lab by Quan... All Bookmarks

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Untitled circuit File Edit View Help Save file View jobs Set up and run

Operations

H	$\oplus$	$\otimes$	$\otimes$	$\otimes$
I	T	S	Z	$T^\dagger$
$S^\dagger$	P	RZ	$\curvearrowright^z$	$ 0\rangle$
$\vdots$	$\bullet$	if	$\sqrt{X}$	$\sqrt{X}^\dagger$

Freeform alignment Inspect

q[0] q[1] q[2] c1

Probabilities

State 000

Probability: 0%

Q-sphere

Phase

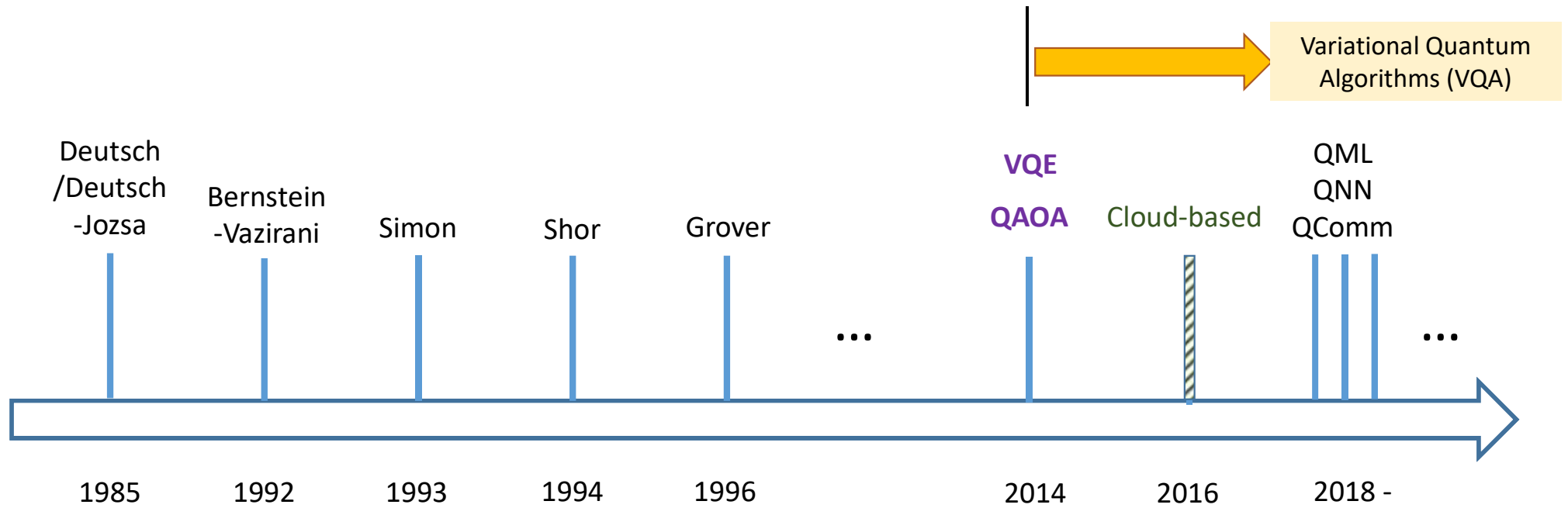
Labels

☒ State ☒ Phase angle

```
1 OPENQASM 2.0;
2 include "qelib1.inc";
3
4 qreg q[3];
5 creg c[1];
6
7 x q[0];
8
9 // @columns [0]
10
```

Quantum Algorithms

Development of Quantum Algorithms



QAOA: Quantum Approximate Optimization Algorithm

Variational quantum algorithms, M. Cerezo et. al., Nature Review Physics, Vol 3, 625-644 (2021)

Function Query – Classical Algorithm

For a computing base $b = \{0, 1\}$, we define function $f(x)$ as

“constant function” if

$f(x) = 0$ or $f(x) = 1$, same output for $x \in \{0,1\}$

and

“balanced function” if

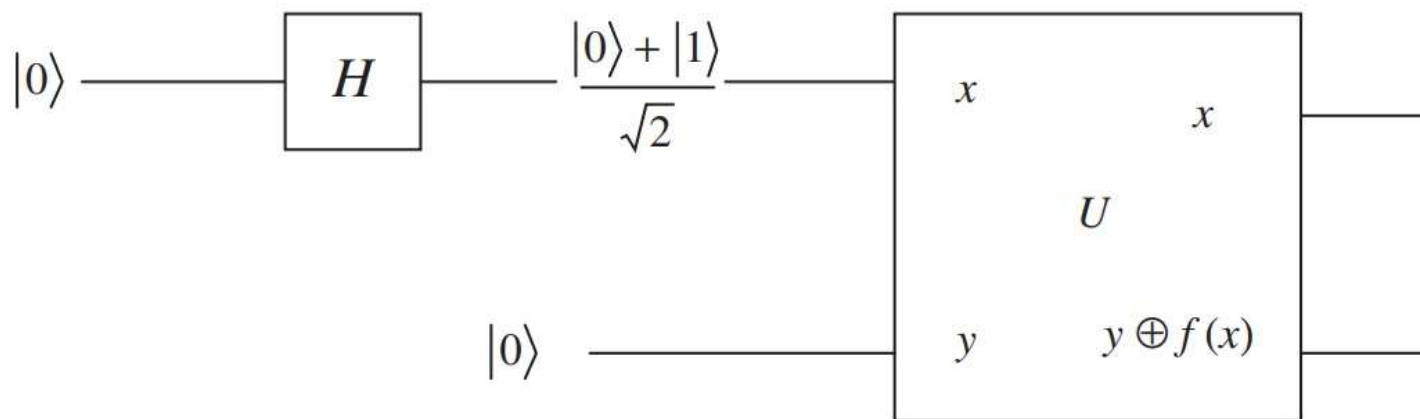
$f(0) = 0; f(1) = 1; f(0) = 1, f(1) = 0$, mixed outputs for $x \in \{0,1\}$

Query: If we select $f(x)$ randomly, how many times we need to invoke the oracle to find what type of function it is?

Can we solve it by just one input?

Deutsch Algorithm – The Quantum Solution

- Start with a two input quantum circuit with inputs x and y such that $x, y \in \{0,1\}$
- Define a unitary function (some logic gate) such that $U|x, y\rangle = |x, y \oplus f(x)\rangle$
- The circuit to implement the algorithm is as follows:

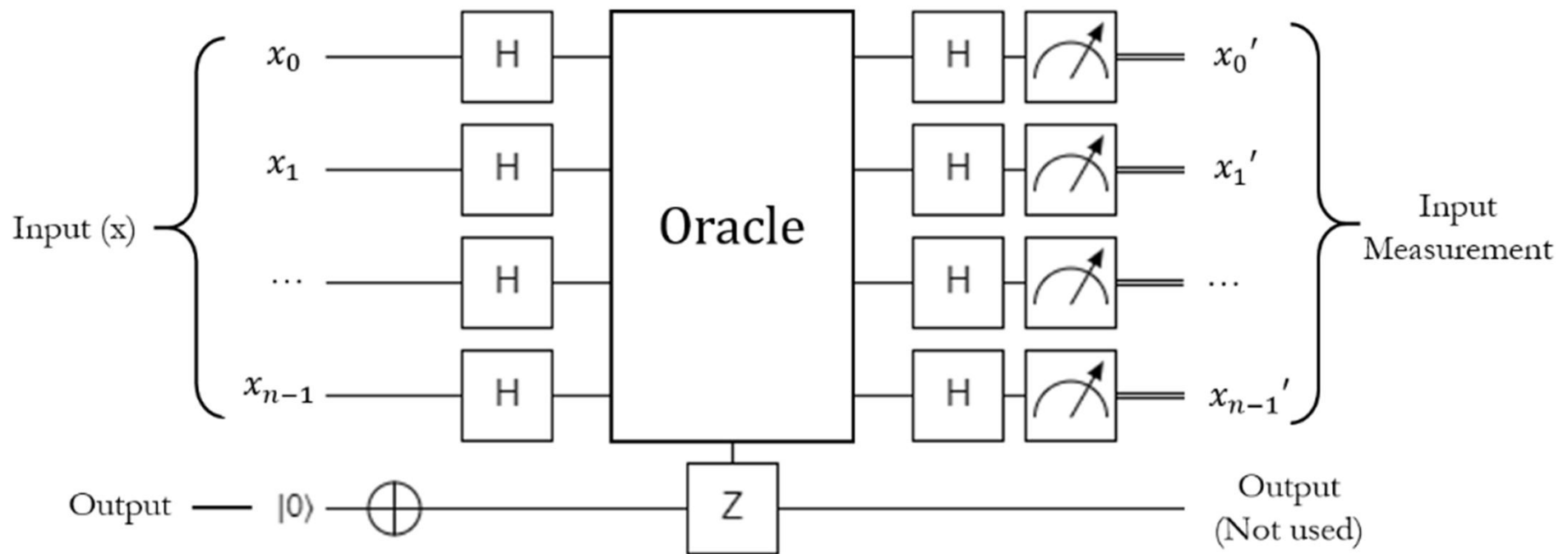


Deutsch Algorithm

- Since $|x\rangle$ is a qubit, it can be in a superposed state
- The action of the unitary function (quantum logic gate) yields:
- $$U\left(\frac{|0\rangle+|1\rangle}{\sqrt{2}}\right)|0\rangle = \frac{1}{\sqrt{2}}U(|00\rangle + |01\rangle) = \frac{1}{\sqrt{2}}(|0, 0\oplus f(0)\rangle + |1, 0\oplus f(1)\rangle)$$
- The final output of the quantum circuit gives:
- $$|\psi_{out}\rangle = (1 - f(0) - f(1)|0\rangle)\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right) + (f(1) - f(0)|1\rangle)\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right)$$
- Check that you can estimate if function is constant or balanced in just one query – Homework!!
- Hint: for constant function, $f(0) = f(1)$ and for balanced function, $f(0) \neq f(1)$

Deutsch-Jozsa Algorithm

- It is an extension of Deutsch algorithm to multiple inputs
- The circuit for Deutsch-Jozsa algorithm is as follows:

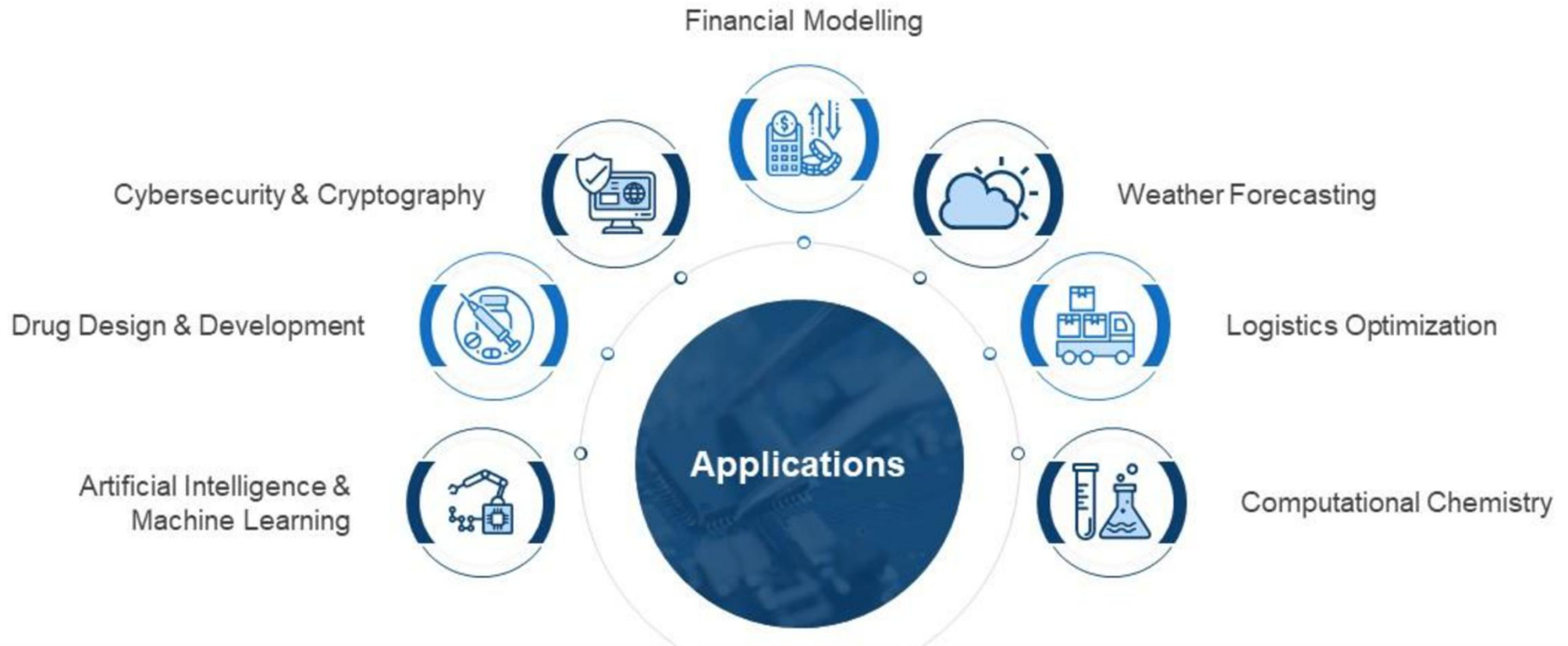


Careers in Quantum Computing

Job Prospects

- Quantum Software Developer – Google, IBM, Samsung, Intel – All want
- Quantum Algorithm Researcher
- Quantum Hardware Engineer
- Quantum Machine-Learning
- Quantum Neural Networks
- Quantum AI
- Quantum Sensing and Quantum Materials
- Quantum Metrology

Areas Where Quantum Computers Will Dominate



Thank You!
Any Question?

